MARRI HARSHA VARDHAN

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PROFESSIONAL SUMMARY

AI Engineering graduate with a strong foundation in Python, SQL, and full-stack development. Completed an AI/ML internship at Tri Tech Innovations, where I helped build an end-to-end AI-powered staffing platform using Flutter for the app interface, FastAPI and PostgreSQL for the backend, and Large Language Models (LLMs) with RAG (Retrieval-Augmented Generation) and MCP (Model Context Protocol) to power intelligent, AI-driven features. Experienced in building machine learning pipelines for NLP (Natural Language Processing) and zero-shot classification techniques that let AI understand and categorize text without needing pre-labeled training data with this research published at an IEEE international conference. Brings a disciplined, production-oriented approach to software engineering, backed by strong practices in API design, database management, version control, and testing.

EXPERIENCE

AI/ML Intern **02/2026 – 06/2026**

Tri Tech Innovations Pvt. Ltd. — Hyderabad

- Designed and developed RESTful APIs (GET, POST, PUT, PATCH) using FastAPI, securing each endpoint with JWT token-based authentication.
- Built and managed relational database schemas using SQLAlchemy and Alembic, including table relationships, foreign key connections, and lookup/reference data across PostgreSQL tables.
- Resolved complex Alembic migration conflicts, including multiple-heads and revision mismatches, ensuring consistent and reliable schema version control.
- Managed Git/GitHub workflows including branch merges, pull requests, and resolution of merge conflicts to maintain clean, production-ready code across team branches.
- Performed unit, integration, and regression testing across APIs and application modules, identified and resolved errors, and ensured functional and non-functional requirements were consistently met.
- Built and deployed Flutter APKs across development, staging, and production environments to support iterative testing, QA validation, and final release cycles.
- Developed Flutter UI screens based on Figma designs using AI-assisted development tools, and integrated them with backend APIs to enable live, end-to-end functionality.
- Collaborated on Chameleon, an AI chatbot scoped to answer job-related candidate queries, using MCP tools to restrict responses and prevent off-topic or sensitive interactions.
- Worked with a small team on refactoring the existing codebase to bring it up to industry standards and make it production-ready.
- Strengthened applied AI/ML knowledge through hands-on sessions on RAG, LangChain, MCP, NLP, and related GenAI concepts, supporting practical implementation across the team's projects.

EDUCATION

B.Tech in Artificial Intelligence **05/2026**

Anurag University — Hyderabad, India **CGPA: 7.2**

Coursework: DSA, Advanced ML, NLP, Big Data Analytics, Database Systems, Cloud Computing, Software Engineering

SKILLS

Programming Languages: HTML, CSS, JavaScript, Python, SQL, Java, Flutter (AI-assisted).

AI & ML: Scikit-learn, TensorFlow, NLP, Transformers, Zero-Shot Learning, RAG Pipelines, Explainable AI (XAI), LLMs, Prompt Engineering, Few-Shot Learning, LangChain, MCP

Technologies: FastAPI, Flask, REST API , Spring Boot, JSON, JWT / Token-Based Authentication, Alembic, SQLAlchemy

Databases: PostgreSQL, MySQL,Vector DB

Tools & DevOps: Git, GitHub, CI/CD, n8n, VS Code, Docker, Postman

PROJECTS

Explainable Usability Classification from App Reviews via Zero-Shot Transformers & RAG LLMs 2026

github.com/MarriHarshaVardhan/Explainable_Usability_Classification...

- Developed an advanced Python application combining zero-shot transformer classification with a RAG pipeline and LLMs to produce explainable, context-aware usability classifications from app reviews.
- Engineered and iteratively refined prompts using chain-of-thought and few-shot prompting to generate accurate, consistent, and contextually relevant LLM explanations.
- Integrated Explainable AI (XAI) techniques enabling non-technical stakeholders to audit model decisions, aligning with responsible AI principles for real-world deployment.

Tech Stack: *Python, Zero-Shot Transformers, RAG, LLMs, XAI, Prompt Engineering*

Zero-Shot Transformer Framework for App Review Usability Classification 2025

github.com/MarriHarshaVardhan/Zero-shot-classification-usability-system-on-app-reviews

- Built a fully automated Python data pipeline that ingests, cleanses, and classifies 2,000+ real-world Google Play Store reviews using zero-shot transformer models — eliminating the need for labeled training data.
- Implemented BART-Large-MNLI and XLM-RoBERTa-XNLI to classify reviews into 8 usability categories; achieved $\kappa = 0.86$ inter-rater agreement and $\rho = 0.68$ correlation with user ratings.
- Applied web scraping, spam filtering, and text normalisation pipeline; built dashboards to visualise sentiment trends and support data-driven product decisions.

Tech Stack: *Python, BART-Large-MNLI, XLM-RoBERTa-XNLI, NLP, Web Scraping, Data Visualisation*

EzStaf – Job Seeker & Recruiter Platform 2026

- Designed and developed RESTful APIs (GET, POST, PUT, PATCH) using FastAPI to power core platform features, including job postings, candidate applications, and recruiter-employer interactions.
- Secured all API endpoints with JWT token-based authentication, ensuring protected access and reliable session management across the platform.
- Used UV as the Python environment and package manager to streamline dependency management and ensure a consistent, reproducible development setup.
- Built and managed relational database schemas using SQLAlchemy as the ORM and Alembic for version-controlled migrations, including table relationships, foreign key connections, and lookup data across PostgreSQL tables.
- Gathered functional and non-functional requirements for job seekers and recruiters to design key features including a job recommendation mode, Candidate matching, an interactive mode, and an AI chatbot.
- Resolved complex Alembic migration conflicts, including multiple-heads and revision mismatches, ensuring smooth and consistent schema evolution as new features were added.
- Designed and developed intuitive, responsive Flutter UI screens based on Figma designs, delivering a consistent cross-platform experience for job seekers and recruiters.
- Executed end-to-end full-stack integration between the Flutter front-end and FastAPI back-end, resulting in a reliable, scalable platform for connecting job seekers with recruiters.

Tech Stack: *Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Flutter*

Chameleon AI Chatbot (AI Assistant for EzStaf Platform) 2026

- Architected a multi-stage LLM pipeline as a REST API, enabling seamless frontend/backend integration; the pipeline includes preprocessing (PII/sensitive data filtering), dual LLM inference calls for intent classification and response generation, and structured post-processing before delivery.
- Engineered a two-pass LLM decision engine: the first call classifies user intent into three routing paths — off-domain queries (static fallback), general job-related queries (direct LLM response), and data-dependent queries (MCP tool invocation) — reducing irrelevant responses and improving response precision.
- Designed and implemented custom MCP tools split across a Job Server and Profile Server, enabling the LLM to dynamically fetch real-time job listings and user profile data from the EzStaf platform database based on inferred query requirements.
- Authored a structured company knowledge base and system prompt that strictly scopes the LLM's context to EzStaf-specific information, preventing hallucinations and off-domain responses from the assistant.
- Built query preprocessing logic to detect and block sensitive or confidential information before LLM inference, adding a security layer to the chatbot pipeline.

Tech Stack: *Python, MCP (Model Context Protocol), LLM Integration, Mistral API*

PUBLICATIONS

A Zero-Shot Transformer Framework for Classifying Google Play Store Reviews to Enhance App Usability
Avinash Pal Lidlaan, A Mallikarjuna Reddy, Rajesh Kolayee, Om Prakash Bhargav Regu, Harshavardhan Marri
2026 International Conference on Communication, Computing and Emerging Technologies (IC3ET) — IEEE
DOI: 10.1109/IC3ET64989.2026.11467449

CERTIFICATIONS

- Google Cloud – Generative AI (2025)
- Microsoft & LinkedIn – Generative AI (2024)
- Cisco – Python Essentials 1 & 2 (2024)
- Cisco – Data Analytics Essentials (2024)
- Cisco – Intro to Cybersecurity (2024)
- CCNA – Enterprise Networking & Security (2024)
- Wipro TalentNext – Java Full Stack (2024)
- Tritech– Artificial Intelligence and Machine Learning Internship (02/2026 - 06/2026)